1 2 3 4 5 6 7

From Question to Story

A binder you build one week at a time: from what you're working on, to a question, a hypothesis, a boundary, and a story you can tell in five minutes

Name

How this binder works

1. **Read first.** Each week opens with one page on the idea, with the examples and sources from class.
2. **Worksheets.** Do them in order, in writing. Each one ends with "Check before you move on": the questions a committee would ask, asked by you first.
3. **Into your binder.** Each week ends with a box. Next week's warm-up starts by copying it. By Week 7, the boxes add up to your talk.

Contents

Week	Topic	What goes into your binder
1	**What does a good PhD look like?**	One sentence, no tool names: what you're working on, and why
2	**Why does anyone care?**	Your three-step sentence: topic, question, significance
3	**What don't you know?**	Your research question
4	**From question to hypothesis**	Your question and hypothesis pair, and the result that tells them apart
5	**Scope: what are you not claiming?**	Scope paragraphs for your question and your hypothesis
6	**Telling a research story**	The one sentence you want your committee to remember
7	**Walk the U: the 5-minute pitch**	A six-slide pitch plan, and your 8-second version after you give it

Three rules that run through every page

Questions, not tasks

A task has no wrong answer. A question can be wrong, and that's what makes it research.

No tool names

The tool is the instrument. Say what you are asking it.

Learned, not did

Lead with what you found. Say what each detail means.

Sources across the binder: Booth et al., *The Craft of Research*; Rugg & Petre, *The Unwritten Rules of PhD Research*; Feibelman, *A PhD Is Not Enough!*; Chamberlin (1890); Judith A. Swan, "Motives in Research"; Ardon Shorr, "Telling Research Stories" (2020); Aurbach et al. (2018).

What Does a Good PhD Look Like?

Problem solvers, not stacks of papers

You already know how to run the tools. This class is about the part that usually goes untaught: choosing the question, and knowing why it matters.

> *…no technical skill is worth more than knowing how to select exciting research projects. Regrettably, this vital ability is almost never taught.*
>
> — Feibelman, *A PhD Is Not Enough!*, p. 2

Problem-oriented, not technique-oriented

TECHNIQUE-ORIENTED

"I'm the person who runs ABAQUS." Your value is the tool. When the tool stops being the right one, so do you.

PROBLEM-ORIENTED

"I'm working out why historic masonry fails under wind load." The tool serves the question. You pick up new methods, or a collaborator, when the problem needs them.

Not a stack of papers

Feibelman's first seven publications were in seven different areas of physics, each on a problem a senior scientist picked for him. Productive and employable, and still, by his own account, a habit that "seriously threatened my career."

The PhD as a master piece

Like an apprentice's cabinet, the dissertation "demonstrates that you have all the skills needed to be a researcher in your own right" (Rugg & Petre, p. 4). The key word is *demonstrates*.

Independent, not supervised

"If you expect your supervisor to know more than you about every aspect of your PhD, then you have missed the whole point" (Rugg & Petre, p. 34). By the end, you should know your problem better than anyone in the room.

What the degree certifies

1. **Identify a problem** worth solving, and not yet solved
2. **Design a way to investigate it:** experiments, not just runs
3. **Communicate what you found:** findings, not activities

The degree is the demonstrated ability to do this cycle on your own.

Sources: Feibelman, *A PhD Is Not Enough!* (pp. 2, 3, 122; Ch. 6); Rugg & Petre, *The Unwritten Rules of PhD Research* (pp. 4, 5, 34).

Are You the Tool or the Problem?

What sets a strong PhD graduate apart from someone who just finished

Start here: Think of someone you'd want to hire, collaborate with, or have on your committee. What can they do that a person who simply completed the requirements can't?

Describe yourself the technique-oriented way. "I'm the person who…" Tool names are fine here, and only here.

Now the problem-oriented way. "I'm working out…" No tool names.

The cycle the degree certifies. Where are you now?

Step	What I can already do on my own	Where I still lean on my advisor
Identify a problem		
Design a way to investigate it		
Communicate what you found		

Check before you move on

- [ ] Would your problem-oriented sentence still be true if you switched tools?
- [ ] Is it something you are trying to find out, or something you do?
- [ ] On the cycle, is the right-hand column longest for "identify a problem"? That is the step this class is for.

WEEK 1 · WORKSHEET 2

What Are You Optimizing For?

Deciding what “good enough” means before the work decides for you

1 2 3 4 5 6 7

Week 1: A good PhD

What does “good enough” look like at the end? Not the minimum to pass. What would you want to be able to do?

What are you optimizing for? Circle any, then rank your top two.

papers a job a skill a question you care about other: _____

What are you *not* optimizing for? What are you willing to do less of?

Look at your last two weeks of work. Which of your answers above did it actually serve?

Check before you move on

- [ ] Is your “good enough” something you could demonstrate, not just something you could list?
- [ ] Does your “not optimizing for” list name something real you will give up?
- [ ] If your work and your answers don’t match, which one needs to change?

Where Is the PhD Taking You?

The PhD opens several doors. It helps to know what's behind each one.

	Tenure-line faculty	Teaching professor	Industry
Primary job	Research, grants, teaching, service	Teaching, curriculum, mentoring	Applied R&D or practice
Evaluated on	Publications, funding, impact	Teaching quality, student outcomes	Deliverables, clients, products
Build during the PhD	Papers, conference talks, grant experience	Teaching portfolio, pedagogy	Technical skills, internships
Tradeoff	Freedom vs. pressure (tenure clock)	Stability vs. less research autonomy	Pay and balance vs. less intellectual freedom
My interest	low 1 2 3 4 5 high	low 1 2 3 4 5 high	low 1 2 3 4 5 high

After Feibelman, *A PhD Is Not Enough!*, Ch. 6, "From Here to Tenure: Choosing a Career Path."

Which path interests you most right now, and why? One paragraph.

What should you be doing during the PhD to keep more than one path open?

Check before you move on

- [ ] Does your paragraph give a reason, not just a preference?
- [ ] Is anything in the "build during the PhD" row for your path missing from your plan this year?

Before You Close the Binder

What you take into next week

Which idea from this week changes how you see your own work?

What is still unclear, or what do you disagree with?

What is one thing you will do differently in the next two weeks?

Into your binder

In one sentence, without naming a tool: what are you working on, and why? Week 2 starts from this sentence.

Come back to Worksheet 2 at the end of the semester. Does your answer to “good enough” still hold?

WEEK 2 · READ FIRST

Why Does Anyone Care?

Turning topics into questions worth asking

1 2 3 4 5 6 7

Week 2: Why care?

Tasks vs. questions

Everything you do is either a task or aimed at answering a question. Both are real, but you need to know which is which.

TASK

Something you do. It has steps and a completion state, but no wrong answer.

"I'm using ABAQUS to model walls."

QUESTION

Something you're trying to find out. It can be wrong, and that's what makes it research.

"Can fire-cut joist retrofits resist EF3 uplift without compromising reversibility?"

The three-step formula

1. **I am studying** ___ names your topic; use action nouns (causes, influence, performance)
2. **because I want to find out** ___ turns it into a question: what you don't know
3. **in order to help my reader understand** ___ answers "So what?" for someone outside your lab

Booth et al., *The Craft of Research*, 3.4.1–3.4.3, pp. 44–45

Worked example

Kallas & Napolitano (2025), *Int. J. Disaster Risk Reduction*

1. I am studying the influence of 3D geometric attributes on blast vulnerability of historic masonry buildings…
2. because I want to find out whether geometric features extracted from 3D models predict damage better than traditional field-survey attributes…
3. in order to help post-disaster responders understand which buildings to prioritize without needing to measure every facade by hand.

Instrumental vs. expressive

INSTRUMENTAL

Behavior that moves you toward your goal. *A student reading six key texts for their area.*

EXPRESSIVE

Behavior that demonstrates effort but produces nothing. *A student reading the same irrelevant page over and over in a state of panic.*

The three-step formula is a diagnostic. If you can fill in all three blanks, your work is instrumental. If you can only fill in Step 1, you might be busy, but not aimed. (Rugg & Petre, *The Unwritten Rules of PhD Research*, Ch. 3)

Also from this week: Heilmeier's Catechism, the reference sheet handed out in session, is on the course Resources page.

WEEK 2 · WORKSHEET 1

1 2 3 4 5 6 7

Week 2: Why care?

What Did You Do, and What Did It Answer?

Sorting your week into tasks and the questions they serve

Warm-up: In one sentence, what are you working on right now, and *why*? No jargon. No tool names. Start from your Week 1 “Into your binder” sentence. If you name a tool, start over.

Your week, sorted. List what you worked on this week. Jargon and tool names are fine here. For each, write the question it helps answer, or “none yet.”

Task (something you did)	Question it serves (something you’re trying to find out)

Check before you move on

- [ ] Does every entry in the right column have a possible wrong answer? If not, it is another task.
- [ ] How many rows say “none yet”? A task with no question isn’t wrong, but you should know which it is.
- [ ] Do several tasks point to the same question? That question is probably your real one.

WEEK 2 · WORKSHEET 2

Your Three-Step Sentence

Topic, question, significance. Up to three research questions, each with all three steps.

1 2 3 4 5 6 7

Week 2: Why care?

> **Rules:** No tool names in Step 1 or Step 3. Step 2 must be something you genuinely don't know yet. Step 3 must name a reader who is not your advisor.

Question 1

1. I am studying	
2. because I want to find out	
3. in order to help my reader understand	

Question 2

1. I am studying	
2. because I want to find out	
3. in order to help my reader understand	

Question 3 (optional)

1. I am studying	
2. because I want to find out	
3. in order to help my reader understand	

Check before you move on

- [ ] Step 1 uses an action noun (causes, influence, performance), not a tool or a method.
- [ ] Step 2 is something you don't know yet, not something you plan to show.
- [ ] Step 3 names a real reader, and says what they would understand that they don't now.

Busy, or Aimed?

Using the three steps as a diagnostic on your own week

Go back to your task list on Worksheet 1. Mark each row **I** (instrumental) or **E** (expressive). What do you notice?

Which of the three steps was hardest to fill in? Why do you think that is?

What would you stop doing, or do less of, if your week only served your three-step sentence?

Into your binder

Your best three-step sentence, revised. Week 3 starts from Step 2.

1. I am studying	
2. because I want to find out	
3. in order to help my reader understand	

WEEK 3 · READ FIRST

What Don't You Know?

Step 2 of your three-step sentence is your research question. This week, make it a good one.

1 2 3 4 5 6 7

Week 3: The question

A research question identifies **what you don't know**. It defines a territory with a range of possible answers.

RESEARCH QUESTION

"What are the structural limits of retrofitting historic URM against tornado loading, and how do those limits interact with preservation constraints?"
Opens a door. Multiple answers are possible. You don't know which one is right.

NOT A RESEARCH QUESTION

"I am optimizing retrofit strategies for fire-cut joist connections using ABAQUS."
That's a task. It tells me what you're doing, not what you're trying to find out.

Is your question real? Three tests

1. **Does it name what you don't know?** If you already know the answer, it's not a question. It's a demonstration.
2. **Does it have more than one possible answer?** "Does CFRP improve capacity?" has only one plausible answer (yes). The interesting question is *how much*, *where it stops helping*, or *what governs the limit*.
3. **Can you imagine evidence that would change the answer?** If no evidence could change your mind, you're not asking. You're asserting.

Common problems with Step 2

Too broad

"I want to find out about tornado damage to buildings." That's a topic. What specifically do you not know?

Foregone conclusion

"I want to find out whether my retrofit improves structural performance." Of course it does. That's why it's called a retrofit. What is the interesting question hiding behind it?

Tool-centric

"I want to find out what DIANA predicts for this loading case." DIANA doesn't predict. You do. What question are you asking the instrument?

Restating Step 1

"I want to find out about the influence of geometric attributes on vulnerability." That's Step 1 with "find out" pasted in front. What do you genuinely not know?

Revisit, Then Diagnose Step 2

Read your sentence back before you change it

Warm-up: Copy your three-step sentence from the Week 2 “Into your binder” box, then read it back to yourself. Does it still sound right?

1. I am studying	
2. because I want to find out	
3. in order to help my reader understand	

Diagnose Step 2. Tick any problem it has, then write the fix.

Problem	Mine?	The fix for mine
Too broad a topic, not a question		
Foregone conclusion of course it does		
Tool-centric names an instrument		
Restating Step 1 “find out about”		

HOW TO FIX EACH ONE

Too broad: narrow it. What specific thing do you not know? **Foregone:** ask how much, where it stops helping, or what governs the limit. **Tool-centric:** remove the tool. What are you asking it? **Restating:** say what you don’t know, not what you’re looking at.

Check before you move on

- [ ] Did you tick at least one box? Most first drafts have one. If you ticked none, read it again as a skeptic.
- [ ] Is the fix a different question, or the same one in new words?

WEEK 3 · WORKSHEET 2

Sharpen It Into a Research Question

Rewrite Step 2 as a question that stands on its own, then test it

1 2 3 4 5 6 7

Week 3: The question

Your research question. A full sentence ending in a question mark. Someone outside your lab should be able to read it without Steps 1 and 3.

Run the three tests. Don't just tick yes. Show the evidence.

Test	Show it
1. It names what you don't know	What, exactly, don't you know yet?
2. It has more than one possible answer	List at least two possible answers:
3. Evidence could change the answer	What evidence would change your mind?

If it failed a test, rewrite it here.

Check before you move on

- [ ] Can it be answered "yes" or "no"? Then it may be a single answer being checked. Week 4 starts there.
- [ ] Do your possible answers in test 2 really differ, or are they the same answer at different sizes?
- [ ] No tool names.

Asking, or Asserting?

What you take into next week

Research questions often come out as statements you already believe with a question mark on the end. That isn't a failure; it means you have a hypothesis. Week 4 separates the two.

Read your question again. Does it ask something, or does it already contain an answer? What makes you say so?

What changed between your Week 2 Step 2 and your question now?

What is still fuzzy about what you don't know?

Into your binder

Your research question, as it stands now. Week 4 starts by reading it back.

WEEK 4 · READ FIRST

1 2 3 4 5 6 7

Week 4: The hypothesis

From Question to Hypothesis

Separating what you don't know from what you think the answer is, and why you need more than one

A hypothesis with a question mark

LOOKS LIKE A QUESTION

"Does the bond between the composite strip and lime-mortar brick govern the capacity of surface-mounted retrofits?"

You already believe the answer (bond) and are asking for "yes."

IS A QUESTION

"*What* governs the capacity of surface-mounted retrofits on historic lime-mortar brick?"

Bond is one possible answer. So is strip strength. So is the masonry itself.

How to tell which one you wrote. Can it be answered yes or no? Do you already expect "yes"? Does it name the answer (a specific mechanism, variable, or direction)? If any is true, you wrote a hypothesis. That's not a failure. You'll need it. It just isn't the question yet.

The move: work backward

"Does bond govern capacity?" → keep it: that's your hypothesis, *bond governs* → write the question it answers: "What governs capacity?" Then list two other possible answers. If you can't, the question is still your hypothesis in different words.

What a hypothesis is

A research question opens a door. A hypothesis walks through it: a **proposed answer**, specific and testable. It can be wrong.

EXAMPLE: POSSIBLE ANSWERS TO ONE QUESTION

A. The bond between strip and wall governs, so a lighter intervention is structurally optimal, which favors preservation. **B.** The strip's own strength governs: more fibre, proportionally more capacity. **C.** The masonry itself fails first, and the retrofit barely engages.

Pick one and commit: A is the hypothesis. B and C are its competitors.

> *Don't set out to collect data in support of your belief. Ask yourself: 'How could I tell if my belief was wrong?' If you can't answer that question, then you're doing politics, not research.*
>
> — Rugg & Petre, *The Unwritten Rules of PhD Research*, p. 215

These are not hypotheses

Unfalsifiable

"My model will provide useful results."

A task in disguise

"I will use ABAQUS to simulate the wall."

Foregone

"The retrofitted wall will perform better."

Too vague

"Building characteristics affect damage."

WEEK 4 · WORKSHEET 1

Question or Hypothesis?

Label what you wrote, and if it's a hypothesis, keep it and write the question above it

1 2 3 4 5 6 7

Week 4: The hypothesis

Warm-up: Copy your research question from the Week 3 "Into your binder" box. Read it to yourself, exactly as written. Does it ask something, or does it already contain an answer?

Run the three checks.

Check	Yes / No
Can it be answered "yes" or "no"?	
Do you already expect "yes"?	
Does it name the answer: a specific mechanism, variable, or direction you already think is right?	

Any "yes" means you wrote a hypothesis. Circle one:

question hypothesis

If it's a hypothesis, write the question it answers. Keep the hypothesis; you'll use it on the next page.

List three possible answers to your question.

My favorite	
A real alternative	
Another real alternative	

Check before you move on: last week's tests

- [ ] The question names what you don't know.
- [ ] It has more than one possible answer, and you just listed three.
- [ ] You can imagine evidence that would change the answer.

WEEK 4 · WORKSHEET 2

1 2 3 4 5 6 7

Week 4: The hypothesis

Propose an Answer, Then Try to Kill It

A hypothesis, what would make you wrong, a competitor, and the experiment between them

Your hypothesis. Pick one of your three answers. Make it specific enough to be wrong.

What would make you wrong? If you can't answer this, you don't have a hypothesis. You have a hope.

It holds if…	It fails if…

EXAMPLE

Holds if the ACI reinforcement efficiency factor collapses to its floor on low-strength masonry, and the FE model shows that connection layout, not strip strength, governs capacity. **Fails if** it does not collapse on low-strength masonry, or the FE model shows that adding more fibre proportionally increases capacity.

A competing hypothesis. One of your other two answers, to the same question.

One experiment that distinguishes them.

The experiment	If your hypothesis is right, you'd see…	If the competitor is right, you'd see…

Check before you move on

- [ ] Your hypothesis isn't unfalsifiable, a task in disguise, a foregone conclusion, or too vague.
- [ ] Both hypotheses answer the same question. If one answers a different question, the pair doesn't hold together yet.
- [ ] The two right-hand cells say different things. If they match, the experiment can't tell them apart.

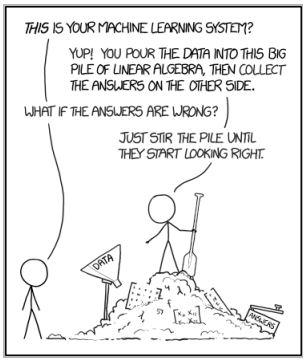
Your Intellectual Child

Why you need more than one hypothesis

> *The moment one has offered an original explanation for a phenomenon which seems satisfactory, that moment affection for his intellectual child springs into existence… There is an unconscious selection and magnifying of the phenomena that fall into harmony with the theory and support it, and an unconscious neglect of those that fail of coincidence.*
>
> — T. C. Chamberlin, "The Method of Multiple Working Hypotheses," *Science* (1890)

If you hold only one hypothesis, you design experiments to confirm it. If you hold two competing hypotheses, you design experiments to distinguish between them. That is a different and better kind of research.

What is your intellectual child? What result would you be tempted to explain away?



xkcd 1838, Randall Munroe

Stir until it looks right. This is the intellectual child with a computer attached. Once you like an answer, it is easy to keep tuning the model, the mesh, or the inputs until the output agrees with you. The pile will always produce an answer. The question is whether you would have accepted a different one.

When did you last stop adjusting because the result looked right, rather than because you had a reason to stop?

WEEK 4 · CARRY FORWARD

Your Question and Hypothesis Pair

Written as a set. This is the deliverable, and Week 5 scopes it.

1 2 3 4 5 6 7

Week 4: The hypothesis

Into your binder

Research question, revised. It asks what you don't know, and it has more than one possible answer.

Hypothesis A: the answer you expect, specific enough to be wrong.

What would make A wrong.

Hypothesis B: another answer to the same question.

The result that would tell A and B apart.

Check before you hand it in

- [ ] Both hypotheses answer the question above them.
- [ ] Neither one names a tool as the answer.
- [ ] You could say what result would make you drop A for B.

WEEK 5 · READ FIRST

Scope: What Are You Not Claiming?

Drawing the boundary around your question and your hypothesis, and getting ready to defend it

Scope is a design decision. It is not a limitation you apologize for. It is the boundary around what your evidence can actually test.

INDEFENSIBLE
Claiming your retrofit works against EF5 tornadoes when your loading represents EF2–3. That isn't ambitious. A committee member will find the gap in one question.

IMPRECISE
Claiming you validated a method when you replicated a published result. That isn't generous. It hides what you actually showed.

> *As paradoxical as it seems, you make your argument stronger and more credible by modestly acknowledging its limits…*
>
> — Booth et al., *The Craft of Research*, §8.3, p. 129

Your question and your hypothesis have different boundaries

Question scope comes from the gap and from who needs the answer. Hypothesis scope comes from where your evidence can tell your answer apart from the competing one. The second is usually smaller. The difference between them is not a weakness: it is where your answer stops and future work starts.

The scope paragraph: four parts

1. **In:** My work covers ___
2. **Out:** It does not cover ___
3. **Why here:** The boundary is here because ___
4. **What sets it:** The evidence or assumption behind that boundary is ___

Every "out" needs a **because**: the material data you have, the assumption your model makes, the loading your code provisions cover. "I didn't have time" is not a reason. "The model assumes X, which stops holding past Y" is.

In the committee meeting

When a committee member asks about something past your line, they usually aren't asking you to do it. They are checking whether you understand what's on the other side. Answer in three moves: **reason** past the line, **anchor** the line, **name the cost**. You don't have to have done it. You have to understand it.

WEEK 5 · WORKSHEET 1

1 2 3 4 5 6 7

Week 5: Scope

Scope Your Question and Your Hypotheses

One column for the question and one for each hypothesis. Read down a column for its scope paragraph.

Start from your Week 4 question and hypothesis pair.

	Research question	Hypothesis A	Hypothesis B
In brief			
In			
Out			
Why here			
What sets it			

EXAMPLE: ONE COLUMN EACH

Question. In: historic URM buildings, any retrofit, tornado loading. Out: modern masonry and new construction; other hazards. Why here: the gap is in historic buildings, where preservation limits which retrofits are allowed. Set by: who needs the answer, owners and engineers of historic buildings.

Hypothesis A. In: surface-mounted composite strips on low-strength lime-mortar brick, within the loading the model represents. Out: cement-mortar masonry, other retrofit families, loading beyond the model. Why here: this is where bond (A) can be told apart from strip strength (B). Set by: the lime-mortar material data and the loading the model represents.

Check before you move on

- [ ] Every “out” has a because, and none of them is “I didn’t have time.”
- [ ] The hypothesis columns are narrower than the question column.
- [ ] Could a committee member push on any one boundary and find you don’t have a reason? Then that boundary isn’t designed yet.

WEEK 5 · WORKSHEET 2

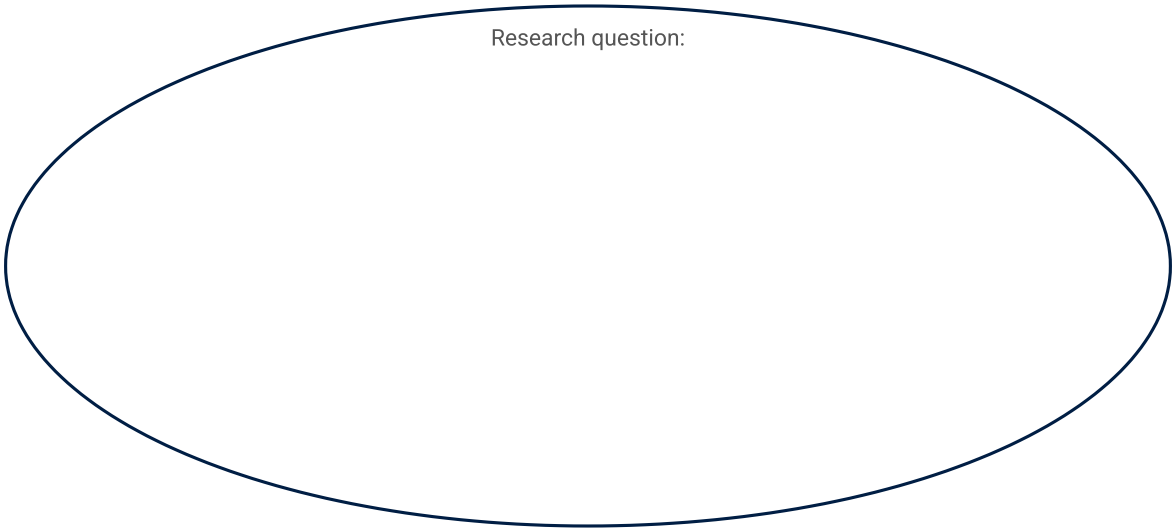
Map Your Boundary

Your question as a bubble, each hypothesis as a circle inside it, and the committee's questions placed where they land

1 2 3 4 **5** 6 7

Week 5: Scope

1. **Label your question.** The big bubble is your research question. Write it in.
2. **Add your hypotheses.** Draw each one as its own circle inside the bubble.
3. **Brainstorm.** Every question a committee member could ask about what's past your lines. Aim for five. Number them below, and put each number on the map where it lands.



Outside the bubble: not this work

Inside the bubble, outside every hypothesis: the question covers it, the evidence doesn't

#	A committee question about what's past your line	Where it lands
1		in a hypothesis · question only · outside
2		in a hypothesis · question only · outside
3		in a hypothesis · question only · outside
4		in a hypothesis · question only · outside
5		in a hypothesis · question only · outside

Check before you move on

- [ ] At least one question lands in "question only." That band is where future work lives.
- [ ] Star the question you would least like to be asked. The next page answers it.

Answer the Hardest One

Show you know the other side of your line without signing up to go there

SHOWS YOU CAN THINK

Predict: "I'd expect… because…" **Mechanism:** "What changes past that point is…" **Literature:** "Others who looked at that found…" **Cost:** "Answering it would take…"

SIGNS YOU UP FOR WORK

"I could add that." "That should be easy to include." "I'll look into running that." "I haven't had time yet."

EXAMPLE: "HOW WOULD YOUR STRIPS DO ON A CEMENT-MORTAR WALL?"

Reason: I'd expect bond to matter less. On stronger masonry the strip has a better substrate to hold onto, so the strip's own strength would start to govern. **Anchor:** That's why I scoped to lime mortar: the bond limit I'm testing shows up on low-strength masonry, and that's the material data I have. **Cost:** Cement mortar would need its own bond tests. It's a natural next study, but it is not part of this dissertation.

The starred question from your map.

Reason past the line what you'd expect there, and why	
Anchor the line why your work stops here	
Name the cost what crossing would take; say it's not this work	

When it really is a request for more work ("You should also…", "Before you defend, I'd want to see…")**: engage the same way, then say "Let me think about how that fits, and follow up with my advisor." Don't agree to new work in the room. Which of your five might be one, and what would you say?**

Check before you move on

- [ ] Most of your answer is about the part you didn't study. That is what shows you know your work.
- [ ] No sentence in it promises to do the work.
- [ ] If the push showed your line has no reason behind it, the line moves. Go back to Worksheet 1.

WEEK 5 · REFLECT AND CARRY FORWARD

You Already Have Most of a Story

Everything you have built so far is a piece of your talk

1 2 3 4 5 6 7

Week 5: Scope

Which boundary are you least sure of, and what would make you surer?

Into your binder

Scope paragraph for your question. In, out, why here, what sets it. No tool names unless the tool sets the boundary.

Scope paragraph for your hypothesis.

The story so far

One line each, copied from your binder. Week 6 puts these in order.

Why anyone cares Week 2, Step 3	
What you don't know Week 3	
Your proposed answer Week 4	
Where it holds this week	
What you found	?

WEEK 6 · READ FIRST

Telling a Research Story

Putting everything you have built into an order an audience can follow

1 2 3 4 5 6 7

Week 6: The story

Invert the pyramid

Papers teach us to establish credibility first: background, then methods, then, finally, the result. A listener stops paying attention long before they get there. **Start with the result and why it matters, then fill in the details.** (After Somerville & Hassol 2011, via Shorr)

Say what the detail means

The detail	What it means
“Excitation was restricted to a femtoliter volume.”	“We eliminated a major source of noise.”
“I’m building an annotated corpus.”	“I’m teaching a computer what to look for.”
“I ran 200 nonlinear analyses.”	? If you can’t fill this in, the analyses are what you *did*, not what you *found*.

Research fits a story template

Goal: the big problem you’re trying to solve
Obstacle: why it’s hard, and why it hasn’t been done
Approach: what you do that gets past the obstacle
Result: what you found (so far)
Benefit: how that result moves the original goal forward

The elements are a checklist. A missing one is a hole in your talk.

“We want ___, but ___”

Chain the obstacles; each one you solve becomes the next, narrower goal. Radiolab on Fritz Haber: the world couldn’t feed a growing population, **but** plants need nitrogen, **but** the nitrogen in the air isn’t in a form plants can use, **but** it clings tightly to itself. Haber used temperature and pressure to turn it into ammonia, and that helps feed billions of people.

The U-shape, and the broken path

A talk starts at the Goal, goes down through the Obstacles, across your Approach, up through your Results, and finishes at the Benefit. The most common failure stays on the bottom row: all Approach, with no Goal it serves and no Benefit it delivers. That is the “here’s what I did” talk instead of the “here’s what I learned” talk.

Adapted from Ardon Shorr, “Telling Research Stories,” Princeton (2020). The Week 7 packet walks the U one stop at a time.

Lead With What You Learned

Turning what you did into what it means

Warm-up: Write your current key finding in one sentence. Read it to yourself. Is this something you *learned*, or something you *did*?

learned did

Say what your details mean. List technical details you'd put in a talk: a method, a dataset, a number of runs, a model choice. Then say what each one means for someone who doesn't do your work.

The detail	What it means

Invert your pyramid. Write the first two sentences of a talk on your work: the result, then why it matters. Background comes later.

Check before you move on

- [ ] Any row you couldn't fill in on the right is something you did, not something you found. Mark it.
- [ ] Your opening sentences would make sense to someone who stopped listening after them.
- [ ] If your warm-up sentence was "did," rewrite it as "learned" using the right-hand column.

WEEK 6 · WORKSHEET 2

Draft Your Storymap

1 2 3 4 5 6 7

Week 6: The story

Fill in all five, one or two sentences each, starting from the pieces already in your binder

Element	Start from	Your draft
Goal	Week 2, Step 3: why anyone cares	
Obstacle	Week 3–4: your research question is the first obstacle	
Approach	Week 4: your hypotheses, and how you'll tell them apart	
Result	New. No result yet? Write what you expect and mark it EXPECTED.	
Benefit	Week 5: it has to stay inside your scope	

Chain your obstacles. Start from your Goal and keep asking "but" until you reach your research question.

We want	
but	
So we want	
but	

Check before you move on

- [ ] Read the Benefit right after the Goal. Does it answer it?
- [ ] Are your Result and Benefit inside the scope paragraph and boundary map from Week 5?
- [ ] Which box is longest? If it's Approach, you're on the broken path.

Where Does Your Story Sag?

What you take into building your pitch

Which element was hardest to write, and why?

Where does your talk drift to the bottom row? What would you cut to get back up?

Look back across your binder. What changed most between Week 2 and now?

Into your binder

The one sentence you want your committee to remember. A finding, or an expected finding marked as expected.

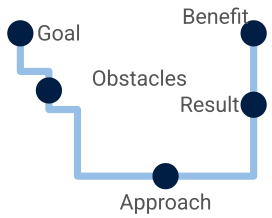
The element you'll work on first. Circle one. The Week 7 packet takes each one in depth.

Goal Obstacles Approach Result Benefit

WEEK 7 · STORYMAP WORKSHEETS

Walk the U, One Stop at a Time

From your Week 6 storymap to a 5-minute pitch you can build in PowerPoint



A talk follows a U: start at the **Goal**, dig down through the **Obstacles**, cross your **Approach**, climb up through your **Result**, and finish at the **Benefit**. These pages take you around it one stop at a time. Do them in order. Each one builds on the page before.

Every page has the same four parts

1. **The question that pulls it out.** If you don't answer it, your committee will ask it.
2. **An example** from the Mayfield talk in Week 6. The setting is real; the numbers are made up for teaching.
3. **Your turn.** Write in sentences, not keywords. Start from what you already have: your Step 3, research question, hypotheses, scope paragraph, and boundary map.
4. **Check before you move on.** These are the questions the group will ask when you pitch. Ask them of yourself first.

The next two pages put the U back together and turn it into a slide plan: six slides, five minutes. The last page is for after you pitch.

Warm-up

Write your current key finding in one sentence.

Is that something you *learned*, or something you *did*? Circle one. If "did," that is fine for now; the Result page is where it gets fixed.

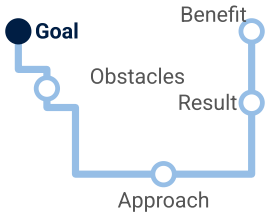
learned did

Adapted from Ardon Shorr, "Telling Research Stories," Princeton (2020); Judith A. Swan, "Motives in Research"; Aurbach et al. (2018).

STAGE 1 OF 5 · TOP LEFT OF THE U

Goal: What Do We Want?

Big picture first. The problem, and who has it.



The question: Why do you want to solve that problem? What happens if you are wildly successful?

EXAMPLE

A preservation office in a tornado town can afford to inspect and retrofit a handful of historic brick buildings a year. **We want a way to pick that handful before the next storm.** Who cares: a town preservation officer. Stakes: a budget and a choice. No literature review.

Who has this problem? Name a person or role who is not your advisor and not another researcher in your area.

What do they need that they don't have? Something concrete: a decision, a number, a design, a warning.

Your Goal in one or two sentences. Try starting with "We want…"

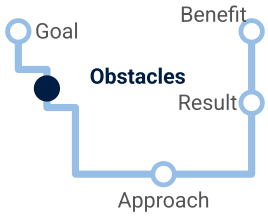
Check before you move on

- [ ] Is the goal clear, and would the person you named say "yes, that's my problem"?
- [ ] Does it match your Step 3 (why anyone cares)? If it changed, which one is right?
- [ ] Could you say it without naming a method or tool? Nobody cares which model yet.

STAGE 2 OF 5 · DOWN THE LEFT ARM

Obstacles: “We Want ___, But ___”

Each obstacle becomes a narrower goal. Keep digging until you hit your research question.



The question: Why is that hard? Why hasn’t it been done? Then ask it again of your answer.

EXAMPLE

We want to pick which historic buildings to retrofit first, **but** a full assessment of every building costs more than a small town can spend. [resources]
So we want a first pass from what you can see from the street, **but** wind design rules were written for modern buildings; nothing says which features of historic brick matter in a tornado. [theory]
Research question: Which features you can see from the street go with more tornado damage in historic brick buildings?

Name each obstacle’s kind. “Nobody has done this” is not an obstacle.

Data: no one has collected it, or not enough

Accuracy: we don’t trust the data or models

Resolution: the data can’t show what is happening

Resources: possible in theory, impractical in time or computing

Inconsistent: plenty of data, but it disagrees

Theory: models are empirical and don’t transfer

Your staircase. Copy your Goal into the first row. Use as many steps as it takes; most people need two or three.

We want		
but		kind:
So we want		
but		kind:
So we want		
but		kind:
Research question		

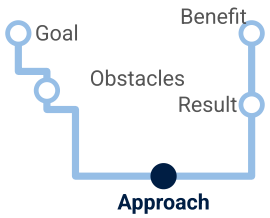
Check before you move on

- [ ] Do the “but”s dig down, each one narrower than the last, to a research question?
- [ ] Is the bottom step the research question you revised in Week 5? If not, which version is better?
- [ ] Does each obstacle have a kind? The kind tells you what approach fits.

STAGE 3 OF 5 · ACROSS THE BOTTOM

Approach: How Do You Tell Them Apart?

Competing hypotheses, and the one comparison that separates them. One slide, not the whole talk.



The question: What do you do? What problem does that solve? (It should solve the bottom obstacle on your staircase.)

EXAMPLE

About 80 historic brick buildings along one EF4 track. **A, geometry:** tall parapets and big openings fail first. **B, location:** it's just distance from the track. **C, hidden condition:** mortar and roof anchorage decide it, not visible from the street.

The test: compare tall and short parapets *at the same distance*. If B is right, the gap disappears. C can't be tested with this data, and the talk says so.

Hypothesis	What it says, in one sentence	If it's right, you would see…
A		
B		
C (if any)		

The comparison that tells them apart. What do you vary, what do you hold fixed, and what do you measure?

What this data can't test. Name it now so your committee doesn't have to.

Check before you move on

- [ ] Are there competing hypotheses, and a way to tell them apart? A list of methods is not an approach.
- [ ] Does each hypothesis predict something different you could actually see?
- [ ] Could you say this on one slide? If you need four, you are on the broken "here's what I did" path.

STAGE 4 OF 5 · UP THE RIGHT ARM

Result: What Did You Learn?

Not what you did. What it means for your hypotheses.



The question: What happened when you did that? Which hypothesis does it favor?

EXAMPLE

Tall-parapet buildings were damaged more at every distance: 85% vs 62% within 100 m, 41% vs 17% past 300 m. **What it means:** distance doesn't erase the parapet gap, so it favors A over B. C is still open.

No result yet? Write what you expect to find and label it **EXPECTED**. That gap is your plan for the rest of the semester.

Say what each detail means.

The detail (what you did or saw)	What it means (what you learned)
"I ran 200 nonlinear analyses."	If you can't fill this side in, it is something you did, not something you found.

Your Result in one or two sentences. Say which hypothesis it supports and which it rules out.

If this were a slide, what would the title say? A sentence that states the finding, not a topic. "Tall-parapet buildings were damaged more at every distance," not "Damage results."

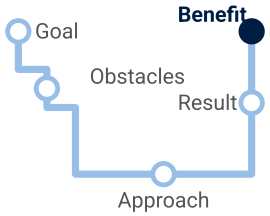
Check before you move on

- [ ] Is there a result (or an expected one, labeled), or only "here's what I did"?
- [ ] Does it answer your research question from the bottom of the staircase?
- [ ] Does it stay inside your scope paragraph and boundary map?

STAGE 5 OF 5 · TOP RIGHT OF THE U

Benefit: Back to the Big Picture

The Benefit answers the question your Goal asked.



The question: What does that result make possible? For the person you named on the Goal page?

EXAMPLE

The Goal asked “which ones first?” **Benefit:** a preservation office can flag parapet height from the sidewalk or aerial photos and spend its inspection budget there.
What this does not tell you: how to retrofit, how to rank two buildings with the same parapet, whether it holds beyond one track. **Next:** inspect connections on flagged buildings to test C.

Copy your Goal here. From Stage 1, word for word.

Your Benefit: how does your Result move that Goal forward?

What this does not tell you. Draw the boundary before the committee asks. Your boundary map has these.

Next step. Often the hypothesis you couldn’t test yet.

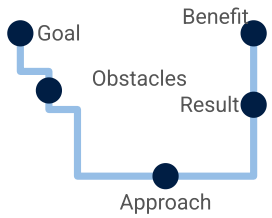
Check before you move on

- [ ] Does the Benefit answer the Goal? Read them one after the other, out loud to yourself.
- [ ] Is the benefit inside your scope, or does it promise more than your Result can carry?
- [ ] Would the person you named on the Goal page know what to do differently?

PUT IT TOGETHER

Your Whole U on One Page

One sentence per stop, copied from the stages. If one won't fit in a sentence, go back to that page.



Goal

Benefit

Obstacles → research question

Result (or EXPECTED)

Approach: hypotheses + the comparison that tells them apart

Is your path broken?

The most common failure is staying on the bottom row: the talk is all Approach, with no Goal it serves and no Benefit it delivers. Look at what you wrote above.

- [ ] All five boxes are filled, and none of them is just a method.
- [ ] Goal and Benefit are about the person you named, not about your model.
- [ ] Approach is the shortest box, not the longest.

The 8-second version

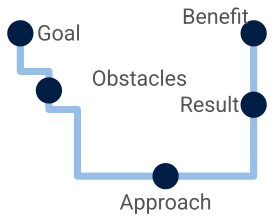
Whatever is left at 8 seconds is your key finding. Example: "In one tornado, tall parapets marked the historic brick buildings that got damaged, even at the same distance from the track."

Compare it with your warm-up sentence. What changed?

TURN IT INTO A PITCH

Six Slides, Five Minutes

Build these in PowerPoint. Every title is a sentence that says something, not a topic.



#	Stop on the U	Slide title (a sentence)	What's on it	Example time
1	**Title = your result**			15 s
2	**Goal**			45 s
3	**Obstacles, down to the research question**			75 s
4	**Approach**			60 s
5	**Result** (or EXPECTED)			75 s
6	**Benefit + what this does not tell you**			30 s

Example times are from the Week 6 Mayfield talk and add to 5:00. Notice that Approach gets one slide.

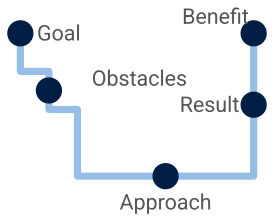
Before you build

- [ ] Read just the six titles in order. Do they tell the story by themselves?
- [ ] Slide 1 states the finding. If someone stops listening there, they still leave with it.
- [ ] No slide is mostly data cleaning, model setup, or tool names.
- [ ] Every chart has a title that says what it means, and a line saying which hypothesis it supports.

AFTER YOU PITCH · REFLECT AND CARRY FORWARD

What Did the Room Hear?

Feedback on your pitch, then half-life your message



Feedback you got, by stop on the U. Write it down as you hear it. Mark the stop that got the most questions.

Goal	
Obstacles	
Approach	
Result	
Benefit	

Half-life your message. Tell your story to a partner. Then again in half the time. Then again. Write what survived each round.

60 seconds	
30 seconds	
15 seconds	
8 seconds	

Aurbach, Prater, Patterson & Zikmund-Fisher (2018). The goal is discovery, not a perfect soundbite.

What did you cut first, and what did you refuse to cut?

Into your binder

Your 8-second version now. Compare it with the one on the “Whole U” page.

The one change you will make before your next talk.